

Hybrid Inheritance

Basic Information

Title: Hybrid Inheritance

File Name: HybridInheritanceDemo.java

Objective / Aim

To demonstrate hybrid inheritance by combining multilevel and hierarchical inheritance structures.

System Requirements

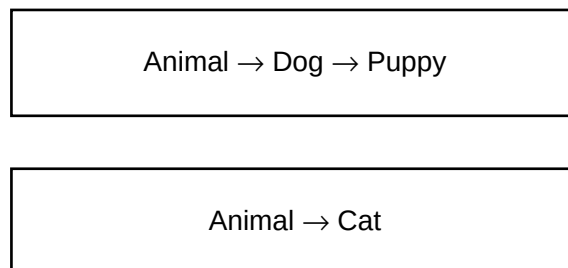
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create Animal as the common parent class.
3. Create Dog extending Animal and Puppy extending Dog.
4. Create Cat extending Animal.
5. Create Puppy and Cat objects.
6. Call the inherited and class-specific methods.
7. End the program.

UML Class Diagram



Source Code

```
// Common parent class
class Animal {

    void eat() {
        System.out.println("Animal is eating");
    }
}

// Dog inherits Animal
class Dog extends Animal {

    void bark() {
        System.out.println("Dog is barking");
    }
}

// Puppy inherits Dog
class Puppy extends Dog {

    void play() {
        System.out.println("Puppy is playing");
    }
}

// Cat also inherits Animal
class Cat extends Animal {

    void meow() {
        System.out.println("Cat is meowing");
    }
}

public class HybridInheritanceDemo {
    public static void main(String[] args) {

        Puppy puppy = new Puppy();
        Cat cat = new Cat();

        // Puppy gets methods from Animal and Dog
        puppy.eat();
        puppy.bark();
        puppy.play();

        // Cat gets method from Animal
        cat.eat();
        cat.meow();
    }
}
```

Compilation & Execution Commands

```
javac HybridInheritanceDemo.java
java HybridInheritanceDemo
```

Output Screen

```
Windows PowerShell
PS C:\Java\Assignment\Assignment1> javac HybridInheritanceDemo.java
PS C:\Java\Assignment\Assignment1> java HybridInheritanceDemo
Animal is eating
Dog is barking
Puppy is playing
Animal is eating
Cat is meowing
```